



EEMG Department
SIMHADRI

Note No :1



Ref:SMTTP/EEMG/2025-26/LMI 243/196416

Date:04/06/2025(11:28:33)

Subject:Revision of LMI for Water audit

LMI Reference No:LMI/OGN/OPS/SYST/034

LMI is revised on the basis of COS-ISO-00-OGN/OPS/SYST/034 Rev. No.: 2 Mar 2025.

Submitted for approval.

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Note No:3

LMI has been revised on the basis of revised OGN dated 02/03/2025.

Revised LMI is in line with the revised OGN.

May please be approved.

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NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

Water Audit

LMI/OGN/OPS/SYST/034 Rev: 03

M A Y -2025

Approved for Implementation by HOP (SIMHADRI)

LMI APPROVAL DOCUMENT

1	LMI NAME	Water Audit
2	LMI No.	LMI/OGN/OPS/SYST/034
3	Reference Document	COS-ISO-00-OGN/OPS/SYST/034 Rev. No.: 2 Mar 2025
4	Superseded Document	LMI/OGN/OPS/SYST/034 Rev 2 dt. March 2023
5	Prepared By	Name: Neeraj Kumar Singh, Sr. Manager (EEMG)
6	Checked By	Name: Navin Kishore AGM (EEMG)
7	Reviewed By	Name: P. K Jena GM (O&M)
8	Approved By	Name: Sameer Sharma H O P (Simhadri)

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TITLE	Water Audit
LMI No.	LMI/OGN/OPS/SYST/034 Rev: 03

1.0 INTRODUCTION

A power station is water extensive industry, large volumes of water is used in various applications as discussed below.

At NTPC Simhadri, water required for steam cooling in condenser is met by water taken from sea (Bay of Bengal) using 3 Nos of pumps having rated flow of 9000 cu.m/hr. at a total head of 40.3 mwc. This sea water is also used for wet disposal of Bottom ash in Ash slurry pump house. The water used in making ash slurry and transportation of this slurry to Ash Dyke is decanted in the ash dyke wells and is recirculated again to the Ash water pump house using Ash water recirculation pumps. A part of the re circulated water is sent to EDP (Effluent Discharge pump house) where, along with Blow down water (CW) and CHP drain it is pumped back into the sea. Flow meters are installed in both the sea water intake line (working and standby) and in the pipeline from EDP to the sea. Sea Water scheme of the plant is enclosed at Annexure-1.

Water requirement for other purposes like PT plant, Fire water is being met by sweet water from YELERU canal. The same is pumped from Sweet water pump house (Stage-I: 4 Nos of pumps: 2 X 750 Cu.mt/Hr. and 2 X 600 Cu. Mt/Hr. Stage-2: 3 Nos of Pumps: 1 X 600 Cu. Mt/hr. and 2 X 750 Cu.Mt/Hr.). From here sweet water is pumped to Raw water reservoir as per the requirement. Through Raw water pumps s w e e t water sent to the plant requirements of PT plant, fire water and service water. The scheme of Sweet water is enclosed at Annexure-2.

Water being a scarce resource all efforts are made to minimize the consumption of water. This objective is achieved through proper selection and design of power plant schemes/systems at initial stage itself and by following proper operation and maintenance practices.

Audit of water balance system also helps in identifying the wastage/inefficient utilization of water. Such audits compare the present flows in different water circuits and compare it with the designed flows, which help in identifying the problematic areas/systems. The audits also should suggest water conservation action plan.

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2.0 Superseded Documents

LMI/OGN/OPS/SYST/034 Rev 2 dt. Feb 2023

3.0 Details of Revision

Entire LMI has been revised according to TC document revision.

4.0 Objective

- 4.1 To carry out third party water audit at Simhadri station in a predetermined time frame.
- 4.2 Scope of work should be clearly defined including usage of STP / ETP / AWRS / Rainwater Harvesting etc. Recommendations for reducing water consumption by exploring ways to reduce evaporation/ leakages / wastages should be the deliverable.
- 4.3 Reducing water consumption by adopting different new technologies should also be suggested. Approval of LMI

5.0 Frequency

Third Party External audit is to be carried out in every 3rd year. Station must carry an internal review of the water consumption and implementation of water audit recommendations. ORT coordinator shall monitor the same. CenPEEP should ensure the stations carry out third party water audit as per frequency decided above.

6.0 SCOPE OF AUDIT & METHODOLOGY

6.1 The following areas need to be covered in the audit:

- Raw water system
- PT water system
- Circulating water (CW) and Cooling tower (CT) system
- CT Blow down
- Fire, service & Drinking water
- Ash Slurry system
- Ash water system & AWRS
- Ash dyke and water reservoir evaporation assessment
- Effluent treatment system (ETP)

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- Sewerage treatment plant (STP)
- Drainage system
- Zero liquid discharge (ZLD)
- FGD system, where made operational.
- Township consumption including horticulture
- Rain water harvesting

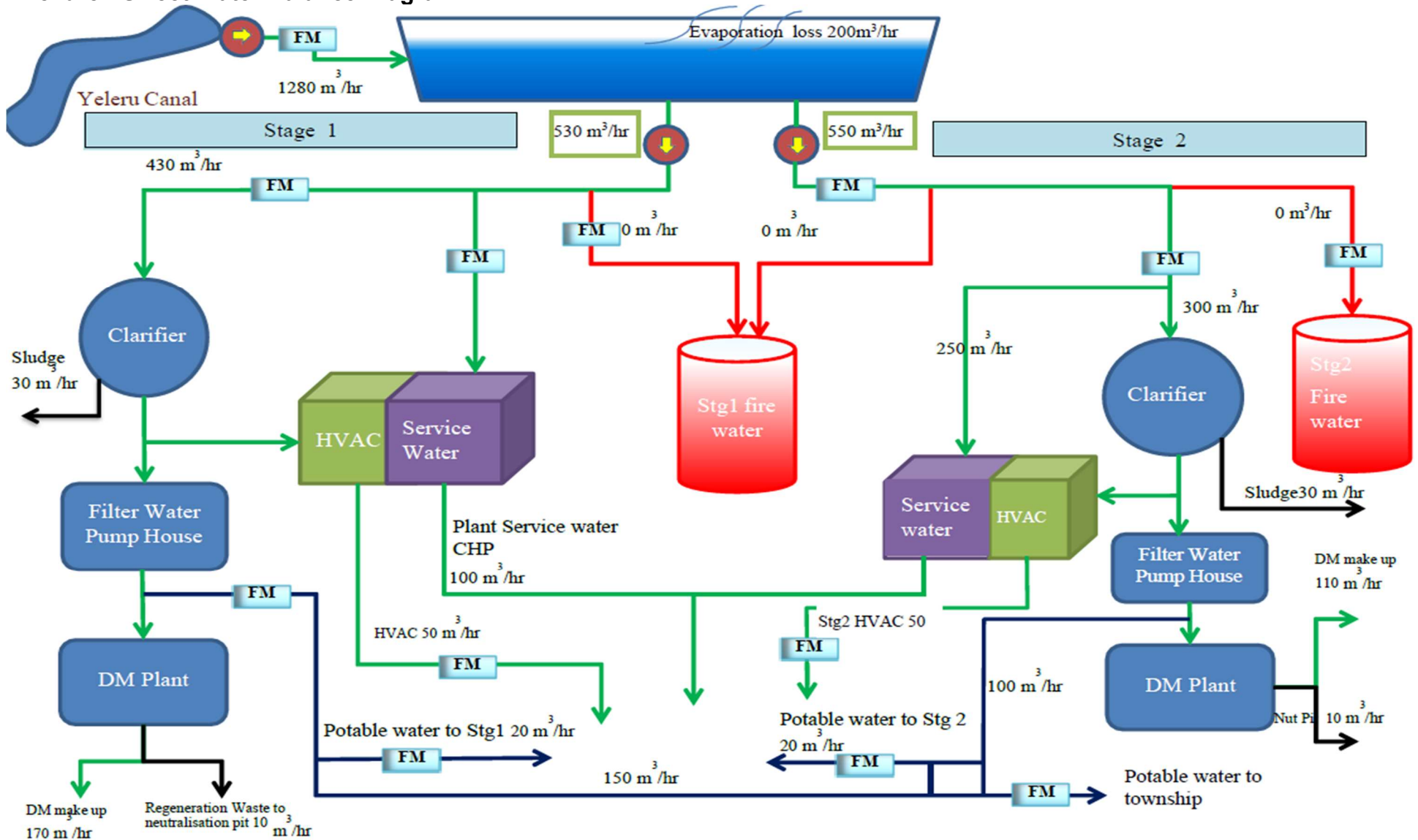
- 6.1** A water balance diagram shall be prepared which will include all consumption and source points.
- 6.2** Portable ultrasonic flow meters are to be used for flow measurements wherever possible. In cases where station flow meters are already installed, a comparison between pre-installed meter and off-line portable meter should be noted. Where it is not technically feasible to use ultrasonic meters, flows are to be estimated following best industry practice.
- 6.3** Cycle of concentration to be checked and evaluated with respect to prescribed COC.
- 6.4** Audit has to necessarily come up with recommendations to conserve/reduce water consumption. It can also suggest successive use of same water in the process keeping its quality in mind. In case where fresh investment is recommended, the same should be supported with cost benefit analysis.
- 6.5** Assessment of power consumption of make up water, raw water and AWRS pumphouse is to be done.

7.0 Review

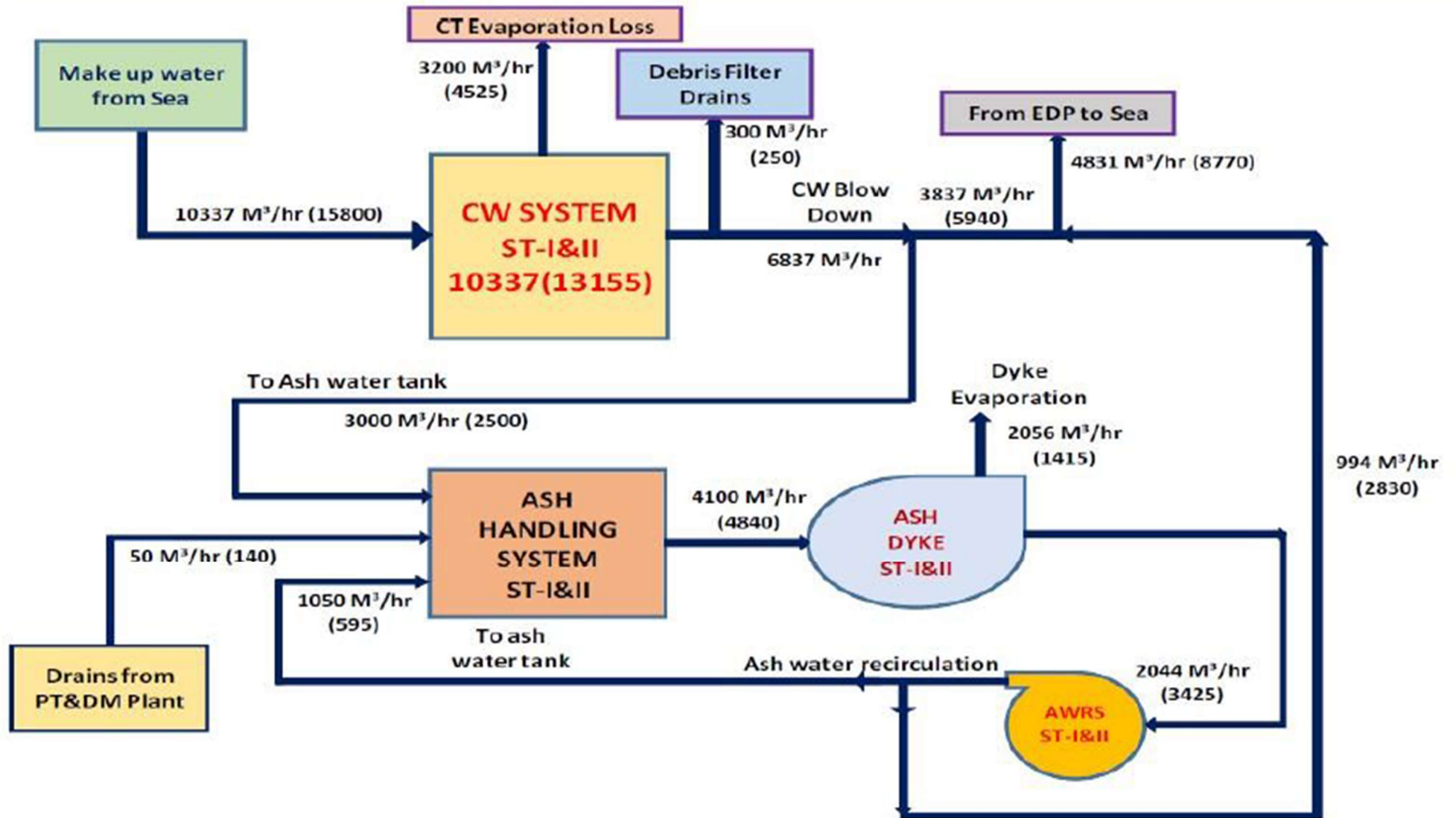
This LMI shall be reviewed by HoP (Simhadri) every 2yrs or as and when necessary.

Annexure-I: Sweet water balance diagram
Annexure-II: Sea water balance diagram
Annexure-III: Checklist

Annexure-I Sweet Water Balance Diagram



Annexure-I Sea Water Balance Diagram



Checklist

1. Summary of Water Saving Measures

	Description	Annual Savings				Total Savings
		Water (M3/Hr)	Power Lakh kWh	Water Rs. Lakhs	Power Rs. Lakhs	Rs. Lakhs
Total Savings						

2. Details of the pumps installed in each subsystem

Sub Sytem:	Eg: - Raw Water / PT Plant / DM Plant/ CW /AHS/ Fire / Service Water etc...				
	Description				Pump 2 Name
1	No. of pumps installed	No.s			
2	No. of pumps operated	No.s			
3	Capacity (Each Pump)	m3/hr			
4	Total Head	m			
5	Motor Rating	kW			

3. Water Balance for each Sub system

Sub Sytem:	Eg: - Raw Water / PT Plant / CW / AHS / Fire / Service Water etc...			
SI No	Description	Inflow (m3/day)	Description	Out flow (m3/day)
1	Pump/Line 1		Pump/Line A	
2	Pump/Line 2		Pump/Line B	
3	Pump/Line 3		Pump/Line C	
4	Pump/Line 4		Other loss like Evaporation/Seepage/ Leakage	
	Total Inflow		Total Out flow	

4. Pump performance for each subsystem

Sub Sytem:	Eg: - Raw Water / PT Plant / CW / AHS / Fire / Service Water etc...					
SI No.	Pump Name	Rated Power (kW)	Design Flow (m3/hr)	Actual Flow (m3/hr)	Actual Power (kW)	Specific Energy Consumption (kW/(m3/hr))
1						
2						
3						
4						

5. Break Up of Specific Water Consumption

SI No.	Sub system	Design Water Consumption as per WBD	Actual Water Consumption	Actual Generation per day	Design Sp Water Consumption as per WBD	Actual Sp Water Consumption
		(m3/day)	(m3/day)	(MWhr)	m3/MWhr	m3/MWhr
1	CT Water consumption (Evaporation + Drift)					
2	Evaporation & Seepage Loss Reservoir					
3	Evaporation & Seepage Loss Ash Dyke					
4	Service Water Consumption					
5	Ash Water Consumption					
6	CHP Water Consumption					
7	FGD Water Consumption					
8	Fire water Consumption					
9	DM Plant Water Consumption					
10	Others					

6. List of diagrams to be prepared

Checklist of Diagrams	
1	Comprehensive Water Balance Diagram
2	Scheme of CW / ACW System
3	Scheme of RW System
4	Scheme of PT plant
5	Scheme of DM Plant
6	Scheme of Ash Water System
7	Scheme of STP
8	Scheme of ETP
9	Scheme of ZLD